

AMEER SALMAN M

Forward Deployed Engineer | Data & AI Platforms

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PROFESSIONAL SUMMARY

Forward Deployed Engineer at Databricks with **8+ years** architecting and delivering enterprise Data & AI platforms inside customer production environments. Builds **agentic AI** systems end to end — multi-agent orchestration, **MCP**-based tool integration, **RAG evaluation**, context optimization and runtime guardrails for autonomous agents — together with the enterprise foundations they depend on: **connectors, identity, data residency and regulatory constraints**. Owns technical delivery from RBI-regulated payment infrastructure at Amazon to AI runtime governance, lakehouse migration and multi-million-dollar workload optimization for Fortune 500 accounts; diagnoses novel, ambiguous failures and turns single-account fixes into open-source and field-wide tooling.

TECHNICAL SKILLS

Python, SQL, PySpark, Spark; **AI/ML** (LLMs, GenAI, Agentic AI, MLOps, Model Deployment, MLflow, Vector Search, RAG, LangChain & LangGraph); **AWS** (Lambda, S3, Redshift, Glue, Athena, Kinesis, Airflow, DynamoDB, IAM, CloudFormation); **Databricks** (Unity Catalog, Delta Lake, DAB, Lakebase, Lakeflow Connect, Genie, Mosaic AI); CI/CD, ETL/ELT Pipelines

PROFESSIONAL EXPERIENCE

DATABRICKS

Forward Deployed Engineer

Remote, India

Jul 2024 - Present

- **Agentic Data Verification (DataPrint)**: Built an end-to-end agentic data verification framework combining an algebraic fingerprinting core with **Foundation Model API (FMAPI)** agents to automate natural-language configuration, dialect drift vs. bug classification, root-cause diagnosis, and self-healing ETL SQL remediation; validated across **16B+ rows × 28 columns** with zero task failures, achieving an **8.1x validation speedup** and **~16,000x network egress reduction**.
- **Autonomous AI Workload Sandbox & Remediation (ShadowClone)**: Architected a production-scale simulation and self-healing framework for the **world's most valuable semiconductor company**, pairing **Delta shallow clones** and **Unity Catalog lineage** with an LLM-driven Autopilot engine (**DSPy** planner + patch executor) to diagnose DAG errors and apply deterministic code/library patches for **DBR and Photon upgrades**; onboarded **~90 enterprise pipelines (1,171 Delta assets)** with **~80% requiring zero code changes**, proving **100% byte-level multiset parity (EXCEPT ALL)** and automated **DBU cost verdicts** with zero production risk.
- **AI Agent Security Gateway & Compliance Engine (AgentGuard)**: Built a runtime governance proxy on **Databricks Apps** to protect enterprise lakehouses from autonomous AI agents and **MCP servers**; integrated a **6-stage checkpoint pipeline** with an **AST SQL firewall** (sqlglot), **LLM intent analysis** and a 6-factor risk engine (0-100 scoring) that blocks **100% of destructive operations** (unbounded DELETE/DROP/TRUNCATE) at **~30-80ms** added latency, delivering **OBO token brokering** and automated lockdown.
- **Agentic RAG Reference Architecture (Carbon-Credit Regulatory Filings)**: Built a retrieval-augmented generation system over the **Verra VCS** registry (**1,500+ Project Design Documents** across 91 countries and 134 methodologies) that answers standards questions and drafts new filings section by section from retrieved precedent — **Docling** semantic chunking with context-enriched embeddings, **hybrid dense + BM25 retrieval** fused by **Reciprocal Rank Fusion** with **cross-encoder reranking**, metadata-filtered **ChromaDB** vector search, and a **LangGraph** agent (intent routing, checkpointed multi-turn memory, streaming) on **Databricks Foundation Model** serving; validated by a **28-test RAGAS-style evaluation suite** spanning faithfulness, context precision/recall, answer relevancy and adversarial robustness.
- **Spark Connector Modernization & Root-Cause Debugging**: Reverse-engineered Spark internals into a maintained **SparkCompat** field fork after Microsoft archived the Spark SQL Server connector, eliminating fatal `NoSuchMethodError` crashes on **DBR 15.4+**, retaining **~15x BulkCopy throughput** over the JDBC fallback and unblocking **Unity Catalog** migrations across **~5 enterprise accounts**.

AMAZON

Data Engineer

Bangalore, India

Oct 2017 - Jul 2024

- **Real-Time Payments Platform**: Migrated batch reporting to real-time ingestion on **DynamoDB Streams**, cross-account S3 replication, **Lambda** and **Glue**, cutting SLAs from **36 hours to 30 minutes**, zero defects over **8+ production months**.
- **Regulatory Migration & Audit**: Led zero-downtime **Redshift**, **QuickSight** and **S3** migrations to **AWS Mumbai (BOM)** for RBI compliance and **GDPR**-compliant EU pipelines; resolved **~₹20B (~\$240M)** in audit exceptions via root-cause analysis.
- **Platform Scale & Automation**: Owned ingestion from **15+ upstream systems** via **120+ ETL jobs**, categorizing **400M+ transactions/month (~4.8B/year)** across four RBI-regulated account structures; automated reconciliation of 20M records (~₹900M knock-offs), cutting **1 week/month** of manual effort and reclaiming **20% Redshift disk space**.
- **Orchestration, CI/CD & Observability**: Orchestrated ETL across **trillions of rows** using **MWAA (Airflow)**, **AWS CDK** and transient **EMR** clusters; built a **Glue CI/CD** framework with **Pytest** data-quality gates plus **CloudWatch** metrics and multi-tier alarms, sustaining **99.9% pipeline uptime** and cutting manual workload **60%**.

EDUCATION

B.E. Mechanical Engineering — Park College of Engineering & Technology, Coimbatore, India | 2017